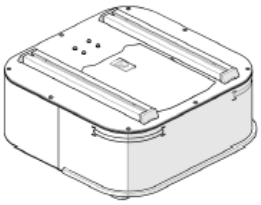

TRACER *MINI*

AgileX Robotics Team

User Manual V1.0.0 2022.05.18

This chapter contains important safety information that must be read and understood by any individual or organization before using the equipment when the robot is powered on for the first time. You can contact us at support@agilex.ai if you have any questions about usage. It is very important that all assembly instructions and guidelines in other chapters of this manual are followed and implemented. Particular attention should be paid to text associated with warning signs.

Safety Information

The information in this manual does not include the design, installation and operation of a complete robotic application, nor does it include any peripherals that may affect the safety of this complete system. The design and use of this complete system requires compliance with the safety requirements established in the standards and specifications of the country where the robot is installed.

It is the responsibility of TRACER MINI's integrators and end customers to ensure compliance with relevant specifications and effective laws and regulations, so as to ensure that there are no major hazards in the complete robot application example. This includes but is not limited to the following:

1. Validity and Responsibility

- Make a risk assessment of the complete robot system.
- Link together the additional safety equipment for other machinery as defined by the risk assessment.。
- Confirm that the design and installation of the peripherals of the complete robot system, including software and hardware systems, are accurate.
- This robot does not have relevant safety functions of a complete autonomous mobile robot, including but not limited to automatic anti-collision, anti-falling, biological approach warning, etc. These functions require integrators and end customers to conduct safety assessments in accordance with relevant rspecifications and effective laws and regulations, so as to ensure that the developed robot does not have any major dangers and safety hazards in practical applications.
- Gather all documents in the technical file: including the risk assessment and this manual.

2. Environment

- When using it for the first time, please read this manual vehicleefully to understand the basic operation contents and operation specifications.
- For remote operation, choose a relatively open area for use, and the vehicle itself does not have any automatic obstacle avoidance sensors.
- Use in an ambient temperature of -10℃~45℃.
- If the IP protection level of the vehicle is not customized separately, the waterproof and dustproof capability of the vehicle is IP22.

3. Inspection

- Make sure that each device has sufficient power.
- Make sure there is no obvious abnormality in the vehicle.
- Check that the remote control's batteries are fully charged

4. Operation

- Ensure that the surrounding area is relatively empty during operation.
- Remote control within sight distance.
- The maximum load of TRACER MINI is 80KG. When in use, make sure the payload does not exceed 80KG.
- When installing an external extension on TRACER MINI, confirm the centroid position of the extension and make sure it is at the center of rotation.
- Please charge in time when the device voltage is lower than 25V.
- When the equipment is abnormal, please stop using it immediately to avoid secondary injury.
- When the equipment is abnormal, please contact the relevant technical personnel, and do not handle it without authorization.
- Please use the equipment in an environment that meets the protection class requirements according to its IP protection class.
- Do not push the vehicle directly.
- When charging, make sure the ambient temperature is greater than 0° C.

5. Maintenance

- In order to ensure the storage capacity of the battery, the battery should be stored with electricity, and it should be charged regularly when not in use for a long time.

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1 Introduction of TRACER MINI n

TRACER MINI is an all-round industrial UGV (UNMANNED GROUND VEHICLE). As a multi-functional modular mobile robot development platform for industrial applications with a modular and intelligent design concept, it has a wide range of applications with strong load capacity and strong power system. The combination of the two-wheel difference and hub motor makes it possible to move flexibly indoors. Stereo camera, lidar, GPS, IMU, manipulator and other equipment can be optionally added to TRACER as an extended application. TRACER MINI can be applied to unmanned inspection, scientific research, logistics and other fields.

1.1 Product list

Name	Quantit
TRACER MINIrobot body	x1
Battery charger 220V	x1
Remote control (optional)	x1
Aviation plug male（4 Pin）	x1
USB to CAN communication module	x1

1.2 Performance parameters

Parameter type	Item	Index
Mechanical parameters	LxWxH(mm)	427.5X 416X 194
	Wheelbase(mm)	269
	Vehicle body weight (Kg)	24~28
	Battery type	Lithium battery 24V 15Ah
	Motor	DC brushless 2 X 150W
	Drive form	Independent drive
	Suspension form	Swing arm independent suspension
	Steering	Differential steering
	Safety equipment	Servo Brake / Bumper
Performance parameters	Idling maximum speed（m/s）	1.8
	Minimum turning radius	Can turn in place
	Code wheel parameters	1024 Line
	Minimum ground clearance (mm)	30
	Maximum gradeability	8°
Control parameters	Control mode	Remote control command mode
	Remote Control	2.4G / critical distance 200m
	Communication interface	CAN

1.3 Required for development

The TRACER MINI can be equipped with a remote control when it leaves the factory. The user can control the TRACER mobile robot chassis through the remote control to complete the movement and rotation operations. Besides, TRACER MINI is equipped with a CAN interface, through which users can conduct secondary development.

2 The Basics

This part will give a basic introduction to the TRACER MINI mobile robot chassis, so that users and developers can have a basic understanding of the TRACER chassis. Figures 2.1 and 2.2 as shown below are overview diagrams of the entire mobile robot chassis.

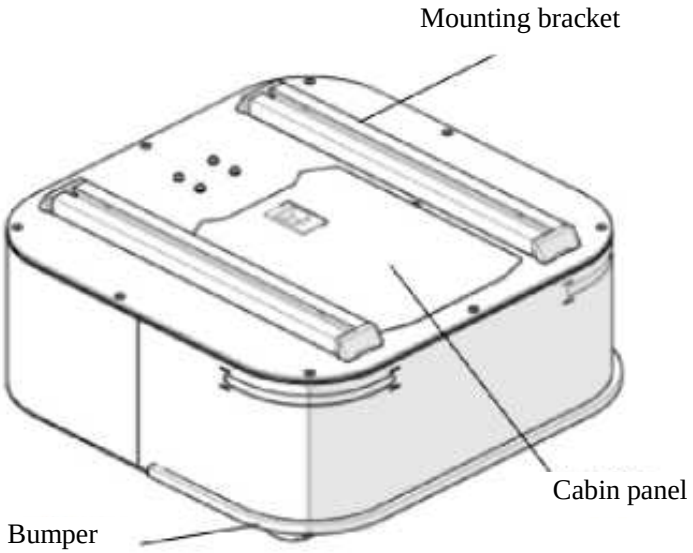


Figure 2.1 Front overview diagram

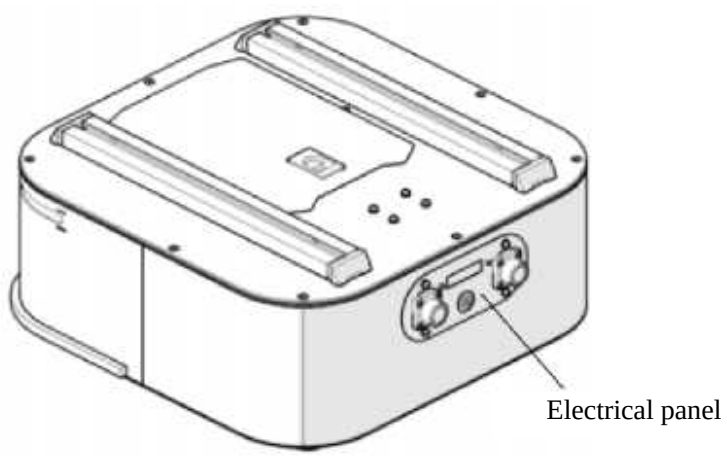


Figure 2.2 Rear overview diagram

TRACER MINI adopts a modular and intelligent design concept as a whole, coupled with a powerful DC hub motor, so that the TRACER MINI robot chassis development platform can move flexibly on the flat ground indoors. The front end of the vehicle body is installed with a bumper, which can cut off the power of the motor in the event of an emergency and reduce the damage to the vehicle body.

Lights are installed in front of the vehicle body, and the front side is designed with blue light for illumination.

The rear of the TRACER MINI is equipped with an open electrical interface and communication interface, which is convenient for customers to vehiclery out secondary development. The electrical interface adopts aviation waterproof connectors in the design and selection, which is conducive to the expansion and use of users on the one hand, and makes the robotplatform available in some harsh environments on the other hand.

2.1 Status indication

Users can determine the status of the vehicle body through the voltmeter, lights and buzzer installed on the TRACER MINI. For details, please refer to Table 2.1.

Status	Description
Current voltage	After power-on, the rear voltmeter lights up, and the current battery voltage can be viewed through the voltmeter in the rear electrical panel, accurate to 1V.
Low voltage alarm	When the battery voltage is lower than 22V, the vehicle body will make a "drip-drip-drip" harsh sound to prompt. When it is detected that the battery voltage is lower than 21.5V, in order to prevent battery damage, TRACER MINI will actively cut off the external extended power supply and actuator power supply. At this time, the chassis will be unable to perform motion control and accept external command control.
Buzzer alarm	For buzzer alarm, one beep (drip...drip....) represents a low-voltage warning. Two beeps indicate that the bumper bar is triggered or the remote control is disconnected. Three beeps represent motor overcurrent. Four beeps means the drive communication is lost. Five beeps represent low-voltage faults. Six beeps represent external 12V overcurrent.

Table 2.1 Vehicle body status description table

2.2 Electrical interface description

2.2.1 Description of rear electrical interface

The extension interface at the rear is shown in Figure 2.3, in which Q1 is the CAN and 24V power extension interface; Q2 is the power display meter; Q3 is the button switch, which is the main switch of the electrical part; Q4 is the power charging port.

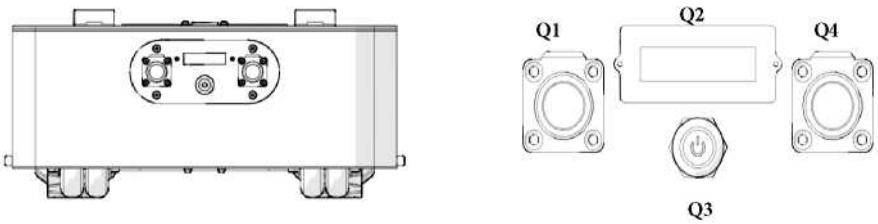


Figure 2.3 Extension interface at the rear

The rear is equipped with a CAN communication interface and a 24V power interface. The specific definition of its line sequence is shown in Figure 2.5.

Pin No.	Pin Type	Function and Definition	Remarks
1	Power	VCC	Positive power supply, voltage range 23~29.2v, maximum current 7A
2	Power	GND	Negative power supply
3	CAN	CAN_H	CAN bus high
4	CAN	CAN_L	CAN bus low

Figure 2.5 Pin description diagram of rear aviation interface

2.3 Remote control instructions

FS remote control is an optional accessory for TRACER MINI products. Customers can choose according to actual needs, and can easily control the chassis of the TRACER MINI universal robot by using the remote control. In this product, we adopt the design of the left-hand accelerator. See Figure 2.6 for its definition and functions.

Vehicle model
Battery voltage
Chassis status
Chassis battery percentage
Parking
Remote control battery
Errors
(Refer to the failure information description table)

Figure 2.6 Schematic diagram of FS remote control buttons

The functions of the button are defined as: SWA and SWD are temporarily disabled. SWB is the control mode selection button, turning to the command control mode when pushed to the top, and the remote control mode when pushed to the middle. SWC is the lamp mode button. S1 is the accelerator button, which controls TRACER to move forward and backward; S2 controls the rotation. POWER is the power button. Press and hold at the same time to turn on the remote control. The back button on the left is used for fault clearing. When pressed, the fault state will be reset. In particular, if the bumper protection is triggered, if the bumper bar is not in contact at this time, press it to reset the fault state; when it is in the contact state, you need to press the button and the accelerator back to release.

2.4 Control command and motion description

We establish the coordinate reference frame of the ground mobile vehicle according to the ISO 8855 standard as shown in Figure 2.7.

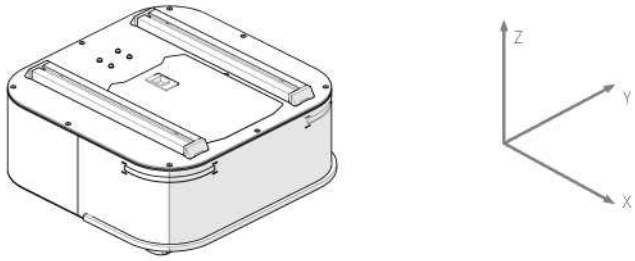


Figure 2.7 Schematic diagram of the vehicle body reference frame

As shown in 2.7, the TRACER MINI body is parallel to the X-axis of the established reference frame.

In the remote control mode, the remote control joystick S1 moves in the positive direction of X when pushed forward, and moves in the negative direction of X when pushed backward. When S1 is pushed to the maximum value, the movement speed in the positive direction of X is the largest, and when pushed to the minimum value, the movement speed in the negative direction of the X direction is the largest. The remote control joystick S2 controls the rotation of the vehicle body left and right. When S2 is pushed to the left, the vehicle body rotates from the positive direction of the X axis to the positive direction of the Y axis. When S2 is pushed to the right, the vehicle body rotates from the positive direction of the X axis to the negative direction of the Y axis. When S2 is pushed to the left to the maximum value, the linear velocity of counterclockwise rotation is the largest, and when it is pushed to the right to the maximum value, the linear velocity of the clockwise rotation is the largest.

In the control command mode, the positive value of the linear velocity means moving in the positive direction of the X-axis, and the negative value of the linear velocity means moving in the negative direction of the X-axis. The positive value of the angular velocity means that the vehicle body moves from the positive direction of the X-axis to the positive direction of the Y-axis, and the negative value of the angular velocity means that the vehicle body moves from the positive direction of the X axis to the negative direction of the Y axis.

3 Getting Started

This part mainly introduces the basic operation and use of the TRACER MINI platform, and introduces how to carry out the secondary development of the vehicle body through the external CAN port and the CAN bus protocol.

3.1 Use and operation

Check

- Check the vehicle body condition. Check whether there is any obvious abnormality in the vehicle body; if so, please contact after-sales support;

Start up

- Press the power switch (Q3 in the electrical panel), under normal circumstances, the voltmeter displays the battery voltage normally, and the front light is on normally;

Power off

- Press the power switch to cut off the power;

Basic operation process of remote control

- After the TRACER MINI robot chassis is started normally, turn on the remote control and select the control mode as the remote control mode, so that the motion of TRACER MINI platform can be controlled by the remote control.

3.2 Charging

TRACER MINI products are equipped with a 10A charger by default, which can meet the charging needs of customers. The specific operation process of charging is as follows

- Make sure that the TRACER MINI chassis is in a power-off state. Before charging, please confirm that Q3 (power switch) in the rear electrical console is turned off;
- Insert the plug of the charger into the Q4 charging interface in the rear electrical control panel;
- Connect the charger to the power supply and turn on the charger switch to enter the charging state.
- When charging by default, there is no indicator light on the chassis. Whether it is charging or not depends on the status indication of the charger

3.3 Development

TRACER MINI products provide a CAN interface for users' development, through which users can command and control the vehicle body.

3.3.1 CAN interface protocol

T TRACER MINI products adopt CAN2.0B standard for the CAN communication standard, with the communication baud rate of 500K and the message format of MOTOROLA. The moving linear velocity and rotating angular velocity of the chassis can be controlled through the external CAN bus interface. Besides, TRACER MINI will feedback the current motion state information and the state information of the TRACER MINI chassis in real time, etc.

The protocol includes a system state feedback frame, a motion control feedback frame, and a control frame. The details of the protocol are as follows:

The system state feedback command includes current vehicle body state feedback, control mode state feedback, battery voltage feedback and fault feedback. The protocol contents are shown in Table 3.1.

Table 3.1 TRACER MINI Chassis State Feedback Frame

Command name	System state feedback command			
Sending node	Receiving Node	ID	Cycle（ms）	Receiving Timeout (ms)
Wire-controlled chassis	Decision control unit	0x211	20ms	None
Data length	0x08			
Location	Function	Data type	Description	
byte [0]	Current vehicle body state	un sig ned i nt8	0x00 System normal 0x02 System exception	
byte [1]	Mode control	unsigned int8	0x00 Standby mode 0x01 Command control mode 0x02 APP control mode (launchedlaterh 0x03 Remote control mode	
byte [2] byte [3]	The upper eight bits of battery voltage The lower eight bits of battery	unsigned int16	Actual voltage X10 (accurate to 0.1V)	
byte [4] byte [5]	The upper eight bits of battery voltage The lower eight bits of battery	unsigned int16	Please refer to the remarks [Fault Information Description]	
byte [6]	Reserved		0X00	
byte [7]	Count check(count)	unsigned int8	0~255 loop count	

Table 3.2 Explanation table of fault information

Fault information description		
byte [4]	bit [0]	Reserved
	bit [1]	External power supply 12V overcurrent (0: no fault 1: fault)
	bit [2]	Reserved
	bit [3]	Reserved
	bit [6]	Reserved
	bit [7]	Bumper protection (0: no fault 1: fault)
byte [5]	bit [0]	Battery undervoltage fault(0: no fault 1: fault)
	bit[l]	Battery undervoltage warning(0: no warning 1: warning)
	bit [2]	Remote control loss protection (0: no fault 1: fault)
	bit(3)	Reserved
	bit [6]	Reserved
	bit [7]	Reserved

The motion control feedback frame command includes the feedback of current vehicle body’s motion linear velocity and motion angular velocity. The specific content of the protocol is shown in Table 3.3.

Table 3.3 Motion Control Feedback Frame

Command name	Motion control feedback command			
Sending Node	Receiving Node	ID	Cycle（ms）	Receiving Timeout (ms)
Wire-controlled chassis	Decision control unit	0x221	20ms	无
Data length	0x08			
Location	Function	Data Type	Description	
byte [0] byte [1]	The upper eight bits of the movement speed The lower eight bits of the movement speed	signed int16	Vehicle body movement speed unit: mm/s	
byte [2] byte [3]	The upper eight bits of the movement speed The lower eight bits of the movement speed	signed int16	Vehicle body rotation angular velocity unit: 0.001rad/s	
byte [4]	Reserved		0x00	
byte [5]	Reserved		0x00	
byte [6]	Reserved		0x00	
byte [7]	Reserved		0x00	

The control frame includes the linear velocity control opening and the angular velocity control opening. The specific protocol content is shown in Table 3.4.

Table 3.4 Motion Control Command Control Frame

Command name	**Control command**			
Sending node	Receiving node	ID	Cycle（ms）	Receiving Timeout (ms)
Decision control unit	Chassis node	0x111	20ms	500ms
Data length	0x08			
Position	Function	Data type	Description	
byte [0] byte [1]	The upper eight bits of the movement speed The lower eight bits of the movement speed	signed int16	Vehicle body movement speed unit mm/s RMS ±1800	
byte [2] byte [3]	The upper eight bits of the movement speed The lower eight bits of the movement speed	signed int16	Vehicle body rotation speed RMS ±1000 units 0.001rad/s	
byte [4]	Reserved	—	0x00	
byte [5]	Reserved	—	0x00	
byte [6]	Reserved	—	0x00	
byte [7]	Reserved	—	0x00	

The light control feedback frame command contains the feedback of the current front light state, as shown in Table 3.5.

Table 3.5 Light Control Command Frame

Command Name	Light Control Frame			
Sending node	Receiving node	ID	Cycle（ms）	Receiving Timeout (ms)
Decision control unit	Chassis node	0x121	25ms	None
Data length	0x08			
Position	Function	Data type	Description	
byte [0]	Light control enabling sign	unsigned int8	0x00 Invalid control command 0x01 Light control enabling	
byte [1]	Front light mode	unsigned int8	0x00 Normally off 0x01 Normally open 0x02 Breathing light mode 0x03 Customer-defined brightness	
byte [2]	Front light custom brightness	unsigned int8	[0,100], where 0 is not bright, 100 is the brightest Note[5]	
byte [3]	Reserved		0x00	
byte [4]	Reserved		0x00	
byte [5]	Reserved		0x00	
byte [6]	Reserved		0x00	
byte [7]	Count check(count)	un sig ned int8	0~255 loop count, count up once every time a command is sen	

Note[5]: This value is valid in custom mode

The light control frame command includes the light control mode and opening. The specific content is shown in Table 3.6.

Table 3.6 Light Control Feedback Frame

Command Name	Light Control Feedback Frame			
Sending Node	Receiving Node	ID	Cycle（ms）	Receiving Timeout (ms)
Wire-controlled chassis	Decision control unit	0x231	20ms	Decision control unit
Data length	0x08			
Location	Function	Data type	Description	
byte [0]	Current light control enabling sign	unsigned int8	0x00 Invalid control command 0x01 Light control enabling	
byte [1]	Front light mode	unsigned int8	0x00 Normally off 0x01 Normally open 0x02 Breathing light mode 0x03 Customer-defined brightness	
byte [2]	Current front light custom brightness	unsigned int8	[0,100], where 0 is not bright, 100 is the brightest	
byte [3]	Reserved		0x00	
byte [4]	Reserved		0x00	
byte [5]	Reserved		0x00	
byte [6]	Reserved		0x00	
byte [7]	Count check(count)	un sig ned int8	0~255 loop count, count up once every time a command is sent	

The control mode frame package is used to set the control mode of the chassis, as shown in Table 3-7

Table 3.7 Control mode setting frame description table

Command name	Control mode setting command			
Sending node	Receiving node	ID	Cycle（ms）	Receiving Timeout (ms)
Wire-controlled chassis	Decision control unit	0x421	None	None
Data length	0x01			
Position	Function	Data type	Description	
byte [0]	Control mode	unsigned int8	0x00 Standby mode 0x01 CAN command control mode[1] Enter standby mode by default after power on	

Note[1] Control mode description

The default control mode is the standby mode, and you need to switch to the command mode to send the motion control command. If the remote control is turned on, it has the highest authority and can block the control of commands. When the remote control switches to the command mode, it still needs to send the control mode setting command before responding to the speed command.

The state setting frame is used to clear system errors, and its specific protocol content is shown in Table 3.8.

Table 3.8 State setting frame description table

Command name	State setting command			
Sending node	Receiving node	ID	Cycle（ms）	Receiving Timeout (ms)
Wire-controlled chassis	Decision control unit	0x441	None	None
Data length	0x01			
Position	Function	Data type	Description	
byte [0]	Control mode	unsigned int8	0x00 Clear all errors 0x01 Clear motor 1 error 0x02 Clear motor 2 error	

Table 3.9 Odometer feedback frame description table

Command name	Odometer feedback command			
Sending node	Receiving node	ID	Cycle（ms）	Receiving Timeout (ms)
Wire-controlled chassis	Decision control unit	0x311	None	None
Data length	0x08			
Position	Function	Data type	Description	
byte [0]	The highest bit of the left wheel odometer	signed int32	Left wheel odometer data Unit mm	
byte [1]	The second highest bit of the left wheel odometer			
byte [2]	The second lowest bit of the left wheel odometer			
byte [3]	he lowest bit of the left wheel odometer			
byte [4]	The highest bit of the right wheel odometer	signed int32-	Rught wheel odometer data Unit mm	
byte [5]	The second highest bit of the right wheel odometer			
byte [6]	The second lowest bit of the right wheel odometer			
byte [7]	The lowest bit of the right wheel odometer			

In addition to the feedback of the state information, the feedback information of the chassis also includes motor information. The following frame feedback is the motor information feedback: in the chassis, the corresponding motor numbers of the two motors are as shown in the following figure:

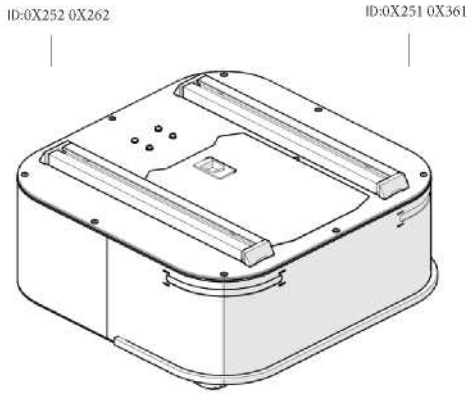


Figure 3.0 Motor Feedback ID Schematic Table

Table 3.10 Motor high-speed information feedback frame

Command name	Motor high-speed information feedback frame			
Sending node	Receiving node	ID	Cycle（ms）	Receiving Timeout (ms)
Decision control unit	Wire-controlled chassis	0x251~0x252	20ms	None
Data length	0x08			
Position	Function	Data type	Description	
byte [0]	The upper eight bits of motor speed	signed int16	Current Motor speed	
byte [1]	The lower eight bits of motor speed		unit RPM	
byte [2]	The upper eight bits of motor current		0x00	
byte [3]	The lower eight bits of motor current	--	0x00	
byte [4]	The current position of the motor is the highest	--	0x00	
byte [5]	The current position of the motor is the second highest	--	0x00	
byte [6]	The current position of the motor is the second lowest		0x00	
byte [7]	The current position of the motor is the lowest		0	

Table 3.11 Motor low-speed information feedback frame

Command name	Motor low-speed information feedback frame			
Sending node	Receiving node	ID	Cycle（ms）	Receiving Timeout (ms)
Decision control unit	Wire-controlled chassis	0x261~0x262	100ms	None
Data length	0x08			
Position	Function	Data type	Description	
byte [0]	Reserve		0x00	
byte [1]	Reserve		0x00	
byte [2]	Reserve		0x00	
byte [3]	Reserve		0x00	
byte [4]	Reserve		0x00	
byte [5]	Actuator state		See Table 3-12 for details	
byte [6]	Reserve		0x00	
byte [7]	Reserve		0	

Table 3.12 Fault information description table

Fault information description		
byte [5]	bit [0]	Reserved
	bit [1]	Reserved
	bit [2]	Whether the motor is overcurrent (0: normal 1: overcurrent)
	bit [3]	Reserved
	bit [4]	Whether the CAN communication is disconnected (0: normal 1: disconnected)
	bit [6]	Reserved
	bit [7]	Reserved

3.3.2 CAN line connection

TRACER MINI provides an aviation plug male as shown in Figure 3.2. For the definition of the line, please refer to Table 3.2.

3.3.3 Realization of CAN command control

Start the TRACER MINI mobile robot chassis normally, turn on the FS remote control, and then switch the control mode to command control, that is, push the SWB mode selection button to the top, then the TRACER chassis will accept the command from the CAN interface, and the host can also analyze the current chassis status through the real-time data fed back by the CAN bus. Refer to the CAN communication protocol for the specific protocol content

Blue: CAN_L
Yellow: CAN_H

Figure 3.2 Schematic diagram of aviation plug male

Note: The power supply in this version can provide a maximum current of 7A.

3.4 Firmware upgrade

In order to facilitate users to upgrade the firmware version of TRACER MINI and bring to customers more perfect experience, TRACER MINI provides the hardware interface for firmware upgrade and the corresponding client software. Its client interface is shown in Figure 3.3.

Upgrade preparation

- Serial X 1
- USB serial portX 1
- TRACER MINI chassis X 1
- PC(WINDOWS operating system) X 1

Firmware upgrade software

- https://github.com/agilexrobotics/agilex_firmware

Upgrade process

- Ensure that the robot chassis power is off before connection;
- Use the Serial to connect to the TRACER MINI chassis to upgrade the serial port, and connect it to the computer;
- Open the client software;
- select the port number;
- TRACER MINIchassis is powered on, click Start Connection immediately, (TRACER MINI chassis will wait for 3S before power-on; if the time exceeds 3S, it will enter the application);if the connection is successful, it will prompt "connection successful" in the text box;
- Load BIN file;
- Click on upgrade and wait for the prompt that the upgrade is complete; Disconnect the Serial, power off the chassis, and power it on again.

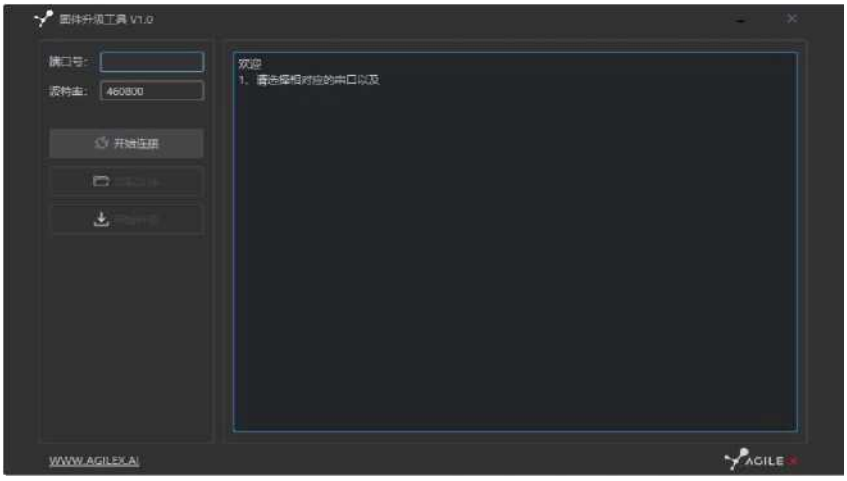


Figure 3.3 Firmware upgrade client interface

3.5 TRACER MINI ROS Package Usage Example

ROS provides some standard operating system services, such as hardware abstraction, low-level device control, implementation of common functions, inter-process messaging, and data packet management. ROS is based on a graphical architecture, so that processes of different nodes can receive, publish, and aggregate various information (such as sensing, control, state, planning, etc.). Currently ROS mainly supports UBUNTU.

Development preparation

Hardware preparation

- CANlight can communication module X1
- Thinkpad E470 Laptop X1
- AGILEX TRACER MINI mobile robot chassis X1
- AGILEX TRACER MINI supporting remote control FS-i6s X1
- AGILEXTRACER MINI top aviation receptacle X1

Environment description of usage example

- Ubuntu 16.04 LTS（this is a beta version, tested on Ubuntu 18.04 LTS）
- ROS Kinetic（also tested in subsequent versions）
- Git

Hardware connection and preparation

- Pull out the CAN line of the TRACER MINI rear aviation plug, and connect the CAN_H and CAN_L in the CAN line to the CAN_TO_USB adapter respectively;
- Turn on the chassis knob switch of the TRACER MINI mobile robot;
- Connect CAN_TO_USB to the USB port of the laptop. The connection diagram is shown in Figure 3.4.

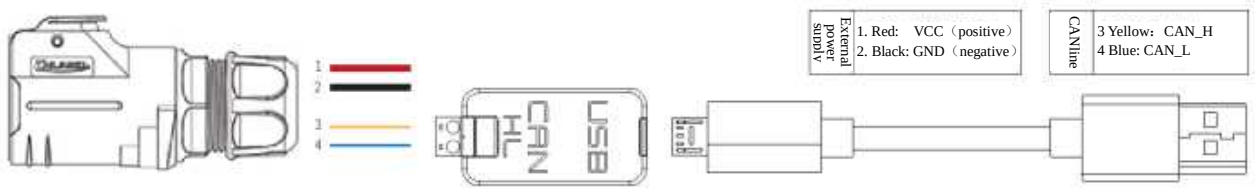


Figure 3.4 CAN line connection diagram

ROS Installation and Environment Setup

For installation details, please refer to http://wiki.ros.org/kinetic/Installa-tion/Ubuntu

Test CANABLE hardware and CAN communication

Set up CAN-TO-USB adapter

- Enabling gs_usb kernel module

  $ sudo modprobe gs_usb
- Set 500k baud rate and Enabling can-to-usb adapter

  $ sudo ip link set can0 up type can bitrate 500000
- If no errors occurred in the previous steps, you should be able to view the can device immediately with the command

  $ ifconfig -a
- Install and use can-utils to test hardware

  $ sudo apt install can-utils
- If the can-to-usb has been connected to the TRACER robot this time, and the vehicle is powered on, use the following commands to monitor the data from the TRACER chassis

  $ candump can0
- Reference：

  [1]https://github.com/agilexrobotics/agx_sdk
  [2]https://wiki.rdu.im/_pages/Notes/Embed-ded-System/Linux/can-bus-in-linux.html

AGILEX TRACER MINI ROS PACKAGE Download and compile

- Download ros dependencies

  $ sudo apt install ros-$ROS_DISTRO-teleop-twist-key- board

  $ sudo apt install libasio-dev
- clone and compile tracer_ros source code

  $ cd ~/catkin_ws/src

  $ git clone https://github.com/agilexrobotics/ugv_sdk.git

  $ git clone https://github.com/agilexrobotics /tracer_ros.git

  $ cd ..

  $ catkin_make

  Reference：https://github.com/agilexrobotics/tracer_ros

Start the ROS node

- Start the basic node through the can port

  $ roslaunch tracer_bringup tracer_robot_base.launch
- Start the keyboard remote operation node

  $ roslaunch tracer_bringup tracer_teleop_keyboard.launch

4 Attention

This part contains some points that should be paid attention to when using and developing TRACER MINI。

4.1 Battery precautions

The battery of the TRACER MINI product is not fully charged when it leaves the factory. The specific battery power can be displayed by the TRACER MINI chassis rear voltage display or read through the CAN bus communication interface; Regarding the charging time, the green indicator light on the charger indicates that the charging is completed, but the battery will still be slowly charged at a current of 0.1A after the green light is on, and it can be charged for about 30 minutes; please do not recharge after the battery is used up. Please charge in time when the TRACER MINI prompts that the battery is low;

Static storage conditions: The best storage temperature is -10℃~45℃. The battery should be charged and discharged once every two months or so when it is not in use, and then the battery should be stored at full voltage. Do not put the battery into fire, or heat the battery, and do not store the battery at high temperature;

Charging: It must be charged with a special charger for lithium batteries. Do not charge the battery below 0° C, and do not use batteries, power supplies and chargers that are not standard in the original factory.

4.2 Precautions for operational environment

The working temperature of TRACER MINI is -10℃~45℃, please do not use it in the environment where the temperature is lower than -10℃ and higher than 45℃;

The relative humidity requirements for the operating environment of the TRACER MINI are: maximum 80%, minimum 30%;

Do not use it in an environment with corrosive or flammable gases or in an environment close to flammable substances;

Do not store around heating elements such as heaters or large coiled resistors; except for the special customized version (IP protection level customized), TRACER MINI does not have water function, please do not use it in the environment with rain, snow and standing water;

It is recommended that the altitude of the environment should not exceed 1000M;

It is recommended that the temperature difference between day and night should not exceed 25℃;

4.3 Precautions for electrical external expansion

The rear expansion power supply current does not exceed

10A, and the total power does not exceed 240W;

When the system detects that the battery voltage is lower than the safe voltage, the external power supply expansion will be automatically cut off, so if the external expansion device involves the storage of important data and there is no power failure protection, users are advised to pay attention.

4.5 Other precautions

When carrying and setting up operation, please do not drop or turn it upside down; for non-professionals, please do not disassemble it without permission.

4.4 Safety precautions

If you have any questions during the use process, please follow the relevant instruction manuals or consult relevant technical personnel;

Before operating the equipment, pay attention to the on-site situation to avoid personnel security problems caused by misoperation;

Do not modify the internal device structure without technical support and permission.

5 Q&A

Q：TRACER MINI starts normally, but the vehicle body does not move with the remote control?

A： First, confirm whether the drive power supply is normal and whether the emergency stop switch is released; then confirm whether the control mode selected by the mode selection switch on the upper left side of the remote control is correct.

Q:When the TRACER MINI remote control is normal, the chassis state and motion information feedback is normal, and the control frame protocol is issued, why the vehicle body control mode cannot be switched, and the chassis does not respond to the control frame protocol?

A: Under normal circumstances, if TRACER MINI can be controlled by the remote control, it means that the chassis motion control is normal, and it can receive the feedback frame of the chassis, which means that the CAN extension link is normal. Please check the sent CAN control frame to see if the command control mode is switched. You can check the status of the error flag in the error bit in the state frame fed back by the chassis.

Q: TRACER makes a "drip-drip-drip..." sound during operation, what should I do?

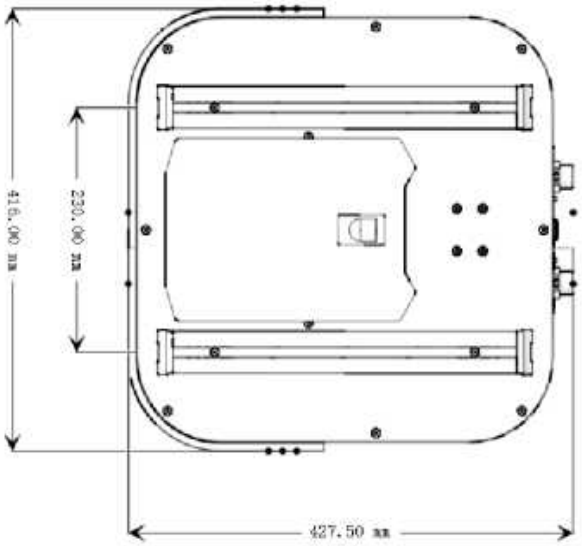
A: If the TRACER MINI emits continuous "drip-drip-drip..." sound, it indicates that the battery is already in the alarm voltage state, please charge it in time; after the relevant sound appears, there may also be an internal related error, you can check the related error through the CAN bus code, or communicate with relevant technical personnel.

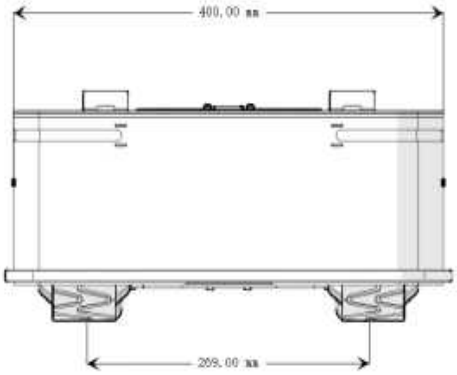
Q:When the relevant communication is carried out through the CAN bus, and the chassis feedback command is normal, why does the vehicle do not respond after the control is issued?

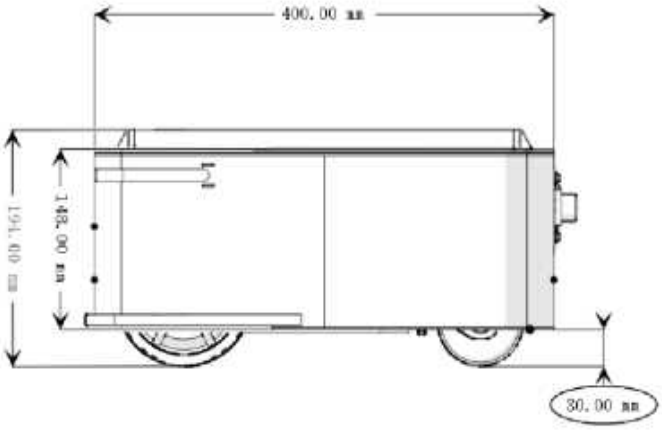
A: TRACER MINI has a communication protection mechanism inside. Chassis has a timeout protection mechanism when dealing with external CAN control commands. Assuming that after the vehicle receives a frame of communication protocol, it does not receive the next frame of control commands for more than 500MS, and it will enter the communication protection with a speed of 0, so the command from the host computer must be periodically issued.

6 Product Dimensions

6.1 Illustrations of product outline dimensions







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